Homework 2

Submission 1, Spring 2026

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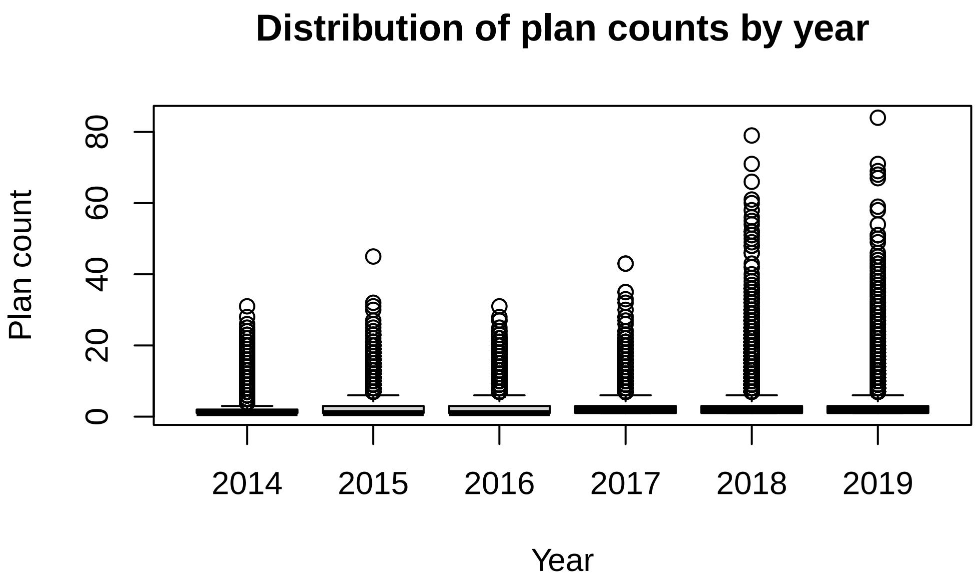
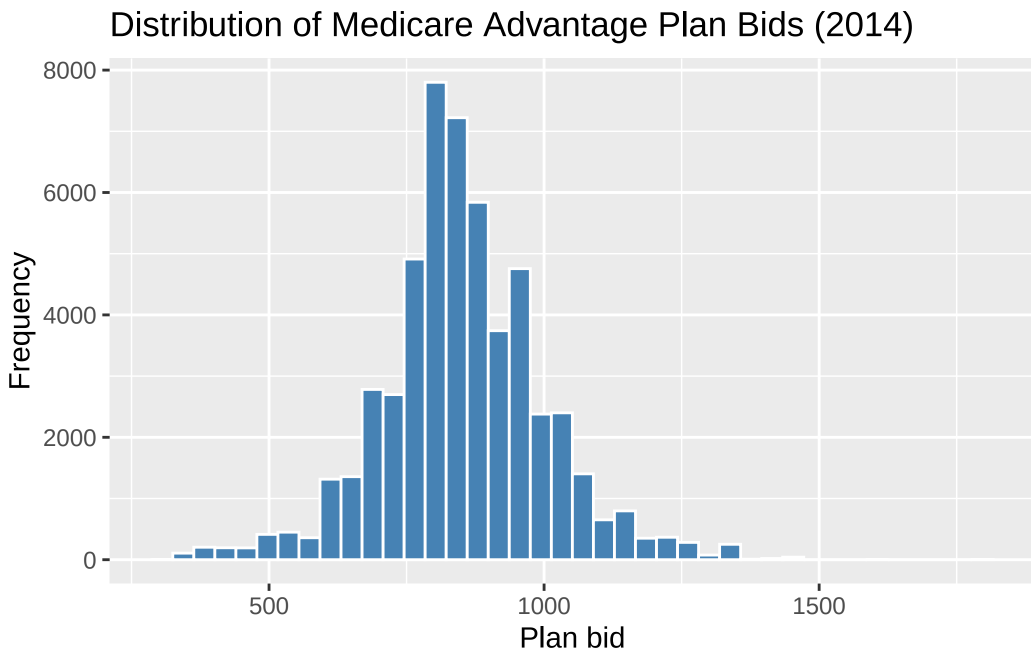
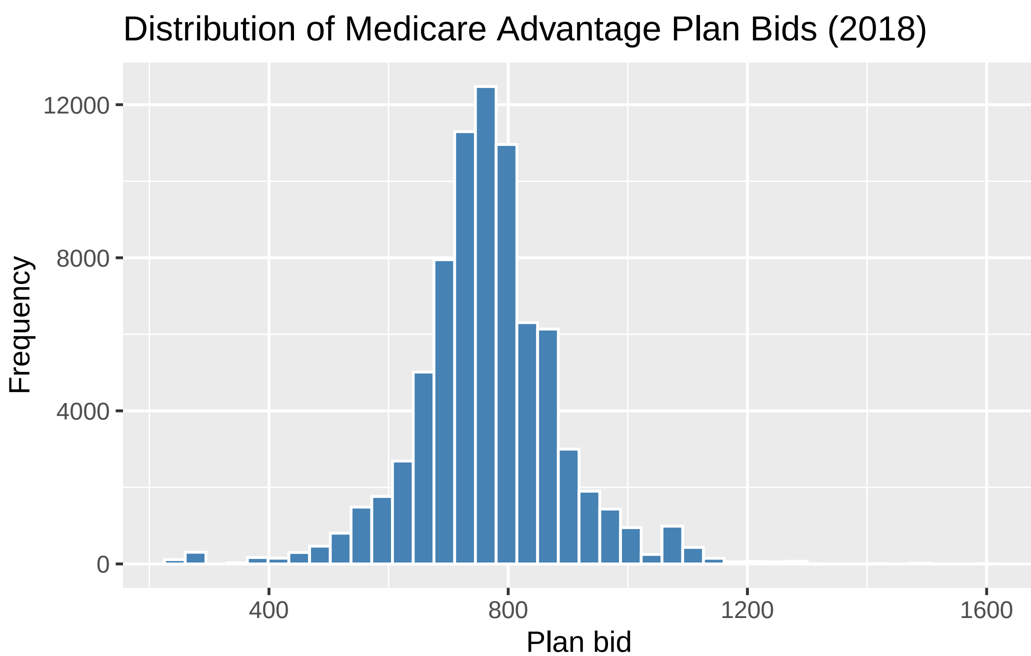
Problem 1





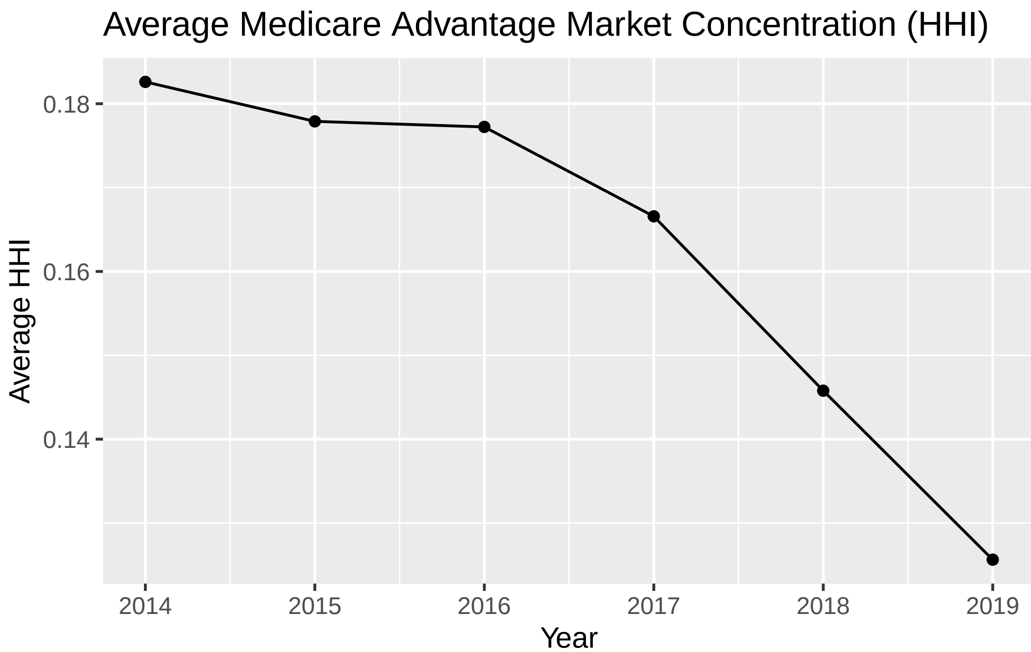


Problem 3

```r
avg_hhi
```

```
# A tibble: 6 x 2
```

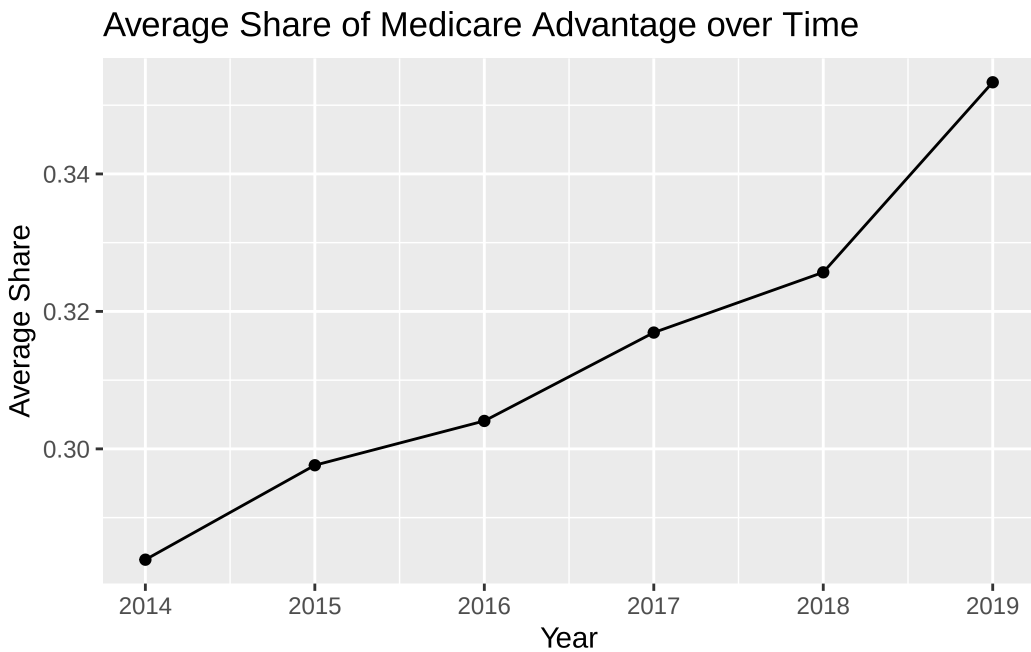
```
  year.x avg_hhi
   <dbl>   <dbl>
1   2014   0.183
2   2015   0.178
3   2016   0.177
4   2017   0.167
5   2018   0.146
6   2019   0.126
```



Problem 4

```
ma_year
```

```
# A tibble: 6 x 3
  year.x  mean groups
   <dbl> <dbl> <chr>
1   2014 0.284 drop
2   2015 0.298 drop
3   2016 0.304 drop
4   2017 0.317 drop
5   2018 0.326 drop
6   2019 0.353 drop
```



Problem 5

```
print(avg_bid_by_market)
```

```
# A tibble: 2 x 2
  market_type   avg_bid
  <chr>           <dbl>
1 Competitive      778.
2 Uncompetitive    771.
```

Problem 6

```
big_avg_table <- bind_rows(q1, q2, q3, q4) %>%
  select(quartile, treated, avg_bid, n)

big_avg_table
```

```
# A tibble: 12 x 4
   quartile treated avg_bid     n
   <chr>    <chr>     <dbl> <int>
 1 Q1       Control    761. 64355
 2 Q1       Treated    779. 21452
 3 Q1       <NA>       824.    59
```

```
 4 Q2       Control    764. 64355
 5 Q2       Treated    770. 21452
 6 Q2       <NA>       824.    59
 7 Q3       Control    768. 64355
 8 Q3       Treated    758. 21452
 9 Q3       <NA>       824.    59
10 Q4       Control    769. 64356
11 Q4       Treated    756. 21451
12 Q4       <NA>       824.    59
```

Problem 7

quartile	ATE
1	-97.16771
2	14.48494
3	41.17377
4	-27.79567

quartile	ATE
1	-97.73672
2	14.56085
3	41.31849
4	-28.11208

quartile	ATE_IPW
1	17.381757
2	5.671065
3	-10.135400
4	-12.776548

quartile	ATE_regression
1	22.646089
2	-8.798631
3	-20.631194
4	-22.646089

Problem 8

Nearest neighbor matching with inverse variance distance and with Mahalanobis distance have very similar average treatment effect values. However, when using inverse propensity weighting and simple linear regression, both methods output noticeably different values.

Problem 9

quartile	ATE_regression_continuous
1	-9.4984154
2	-0.1257976
3	-7.3484978
4	-2.7936440

Problem 10

Working with this data allowed me to see how different estimators will give you vastly different values. One thing that I learned was that it is crucial to know what each name of the row represents so that there is no issue missing a part of the dataset. One thing that surprised me was how easy it was for library or model choices to cause errors or slow runtimes.